GG-B-00406b(DSA-DM)
August 12, 1970
INTERIM REVISION OF
Fed. Spec. GG-B-406a
January 28, 1958

INTERIM FEDERAL SPECIFICATION

BLADES AND FRAME, DENTAL LABORATORY SAW

This Interim Federal Specification was developed by the Defense Personnel Support Center, 2800 South 20th Street, Philadelphia, Pennsylvania 19101, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above. The General Services Administration has authorized the use of this Interim Federal Specification as a valid exception to Federal Specification GG-B-406a.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers an adjustable saw frame and blades suitable for dental laboratory use.

1.2 Classification. This specification covers the following types and classes of items as specified (see 6.1).

Type I. Saw frame.
Type II. Blade.

Class 1. Flat blade.
Class 2. Spiral blade.

2. APPLICABLE DOCUMENTS

2.1 Specifications, standards, and drawing. The following specifications, standards, and drawing, of the issue in effect on the date of invitations for bid or requests for proposal, form a part of this specification to the extent specified herein:

FSC 6520

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SPECIFICATIONS

Federal

QQ-Z-363	Zinc-base Alloy, Die Castings.
MMM-A-250	Adhesive, Water-Resistant, (For Sealing Fiberboard Boxes).
PPP-B-566	Boxes, Folding, Paperboard.
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Boxes, Fiberboard.
PPP-B-676	Box, Setup.
PPP-T-76	Tape, Pressure-Sensitive Adhesive Paper, (For Carton Sealing).

STANDARDS

Federal

FED-STD-123	Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston; New York; Atlanta; Chicago; Kansas City, Mo.; Fort Worth; Denver; San Francisco; Auburn, Washington; and Washington, D.C.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

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DRAWING

Defense Personnel Support Center (DPSC)

Drawing 20537 Frame, Dental Laboratory Saw.

(Copies of Military standards, and drawing required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. Material shall be as specified below, shall be of the highest quality, and shall be suitable in every respect for the intended purpose.

3.1.1 Steel.

3.1.1.1 Type I, saw frame.

3.1.1.1.1 Bow. The saw frame bow shall be fabricated of nickel-plated carbon steel or chromium-over-nickel-plated carbon steel.

3.1.1.1.2 Bearing surface. The blade bearing surfaces (attached to zinc castings) of the saw frame shall be fabricated of nickel-plated carbon steel or chromium-over-nickel-plated, heat treated, carbon steel.

3.1.1.2 Type II, blades. Classes 1 and 2 saw blades shall be fabricated of heat treated steel having a carbon content of 0.7 to 0.8 percent. Blades shall be treated to retard corrosion.

3.1.2 Zinc alloy. The bow support and tang, and the blade holding device at the end of bow of the saw frame shall be zinc-base alloy, die castings, fabricated of alloy AG40A having 0.25 percent maximum copper, and conforming to QQ-Z-363.

3.1.3 Hardwood. The handle of the saw frame shall be fabricated from suitable, seasoned, close-grained hardwood.

3.2 Style, design, and dimensions. The style, design, and dimensions of the type I saw frame shall conform to Defense Personnel Support Center (DPSC) Drawing 20537. Type II saw blade dimensions shall be as specified in table I. The saw (frame plus blade) shall be suitable for cutting plastics and gypsum products in the dental laboratory.

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Table I. Dimension of blades

Class	Length[2] (inches)	Length of Cut (min.) (inches)	Thickness[1]			Blade[2] Width (inch)
			Back (inch)	Ends (inch)	Over Teeth (inch)	
1	5	3.5	0.011	--	0.013	0.022
2	5	3.1	---	0.016	---	0.035

[1]Plus or minus 0.001 tolerance allowable.

[2]Plus or minus 5 percent allowable.

3.3 Construction.

3.3.1 Type I, saw frame. The type I saw frame shall be easily adjustable and shall be suitable for use with either class 1 or class 2 saw blades. The frame shall consist of a bow, bow support and tang, and handle. The frame shall securely hold the blade and shall maintain rigidity during use. All parts of the frame shall be held together when the blade is not in place. The blade bearing surfaces of the holding devices on the rod and tang shall be serrated to provide gripping surfaces for the blade. The handle shall be securely attached to the tang by means of a heavy duty ferrule.

3.3.2 Type II, blades. Class 1 and class 2 saw blades shall be suitable for cleanly cutting through hardened, high strength, dental stone, hardened plaster of paris, or plastic.

3.3.2.1 Class 1, flat blade. Blades shall have a single cutting edge, and shall have blank ends for clamping into the holding devices of the frame. The cutting length of the blades shall have 13 to 15 teeth per inch.

3.3.2.2 Class 2, spiral blade. Blades shall be spiral in design so as to cut in any direction without turning the frame. The cutting length of the blades shall be twisted a minimum of 8 turns per inch, and shall have 33 to 39 teeth per inch. The blades shall have blank ends for clamping into the holding devices of the frame.

3.4 Finish (Type I). The bow and blade bearing surfaces of the holding devices shall have a polished finish. The die cast bow support and tang and bow holding device shall have a smooth, as cast, finish. The handle shall have a smooth, painted or lacquered finish.

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3.5 Identification marking. The saw frame shall be permanently and legibly marked with the name or registered trademark of the manufacturer. Instruments of foreign (other than U.S.) manufacture shall also be marked with the name of the country of origin.

3.6 Workmanship. Workmanship shall be first class throughout. The saw blades and frame shall be free from defects which detract from their appearance or may impair their serviceability, proper functioning, or intended use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties).

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the metals, may be furnished in lieu of actual performance of such testing by the contractor, provided that lot identity has been maintained and can be demonstrated to the Government. The Certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the FSN, the item identification, the name of the material, the lot number, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

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4.2 Quality conformance inspection. The examinations required to assure conformance of those items or lots of items to be offered are classified as specified in tables II and III (see 4.2.1 and 4.3).

4.2.1 Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. The inspection levels and acceptable quality levels (AQLs) shall be as indicated in table II. Unit of product for examination purposes shall be one saw frame (type I), or one package (12 blades) of blades (type II).

Table II. Examination Criteria

Type defect	Inspection level	AQL (percent defective)
Major A	II	1.0
Major B	S-2	1.5
Minor A	II	2.5
Minor B	S-2	4.0

4.3 Examination. The laboratory saw frames and blades shall be examined to determine compliance with the requirements contained in this specification. Examination shall be conducted as specified in 4.2.1. Any saw frames or blades in the sample containing one or more defects shall be rejected, and if the number of defective saw frames or blades in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected. The saw frames or blades shall be examined for defects including, but not limited to, those listed in table III.

Table III. Classification of defects

Category	Defect
Major A:	
101	Style and design of saw frame or blades not as specified.
102	Saw frames not free of sharp edges, cracks, or burrs.
103	Construction of saw frames not as specified.
104	Saw blades are not securely and rigidly held in saw frames.

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Table III. Classification of defects (Cont'd)

Category	Defect
Major B:	
151	Number of teeth per inch for the cutting length of classes 1 or 2 saw blades not as specified.
152	Number of twists per inch for the cutting length of class 2 saw blades not as specified.
Minor A:	
201	Finish of saw frames not as specified.
202	Saw frame or blades not free from dirt, polishing compound, or foreign matter.
203	Bearing surfaces for holding saw blades not serrated.
204	Identification marking not as specified.
205	Marking on packaging and pack not as specified.
206	Packaging and packing not as specified.
Minor B:	
251	Dimensions of saw frames or blades not as specified.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, or C, as specified (see 6.1).

5.1.1 Unit of issue. One type I saw frame or one package of type II, class 2 blades (12 blades), as specified, constitutes one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in table IV shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in the procurement document.

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Table IV. Packaging Quantities

Item	Unit package	Intermediate package	Exterior container
Saw frame (Type I)	1 unit	6 units	96 units
Blade (Type II, class 2)	1 unit	12 units	288 units

5.1.2.1 Variation permitted. If the required number of units in the entire shipment is less than the number of units specified to be overpacked in an exterior container, such units may be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

5.1.3 Level A.

5.1.3.1 Unit package. Unit package shall be a box of appropriate size, constructed in accordance with PPP-B-566, or PPP-B-676, except that commercial colors will be acceptable. Closure shall be adequate to prevent accidental opening.

5.1.3.2 Intermediate package. Intermediate package shall be a box of appropriate size and design, constructed in accordance with PPP-B-636, type CF, class weather-resistant. Boxes shall be closed by taping the top and bottom flaps with 2-inch wide tape conforming to PPP-T-76. Closing of bottom flaps with either adhesive (conforming to MMM-A-250), or metal stitches is permitted.

5.1.4 Level C. Units shall be packaged in standard commercial containers, of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.1).

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for a type 2 load and shall be constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant, grade V3. Closure and strapping shall be as specified in the appendix of the applicable box specification.

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5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall be designed for a type 2 load and shall be constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers, of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Civil agencies. Marking for shipment shall be in accordance with FED-STD-123.

5.3.2 Military activities.

5.3.2.1 Interior packages. Each unit package and each intermediate package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number and name of contractor shall be shown.

5.3.2.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type and class required (see 1.2.1).
(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

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6.2 Specification changes. Because of the extensiveness of changes to this document, the margins have not been marked with an asterisk to indicate where changes from the previous issue have been made.

6.3 Limitations. This specification does not cover all types, classes, grades, styles, or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover those items which are normally procured to meet Federal Government requirements.

Custodians:

Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Review activities:

Army - MD
Navy - MS
Air Force - 03

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

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CAUTION

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DO NOT CONDITION OR BASE YOUR BID/OFFER ON ANY CURRENT PROCUREMENT ON THE INFORMATION SUBMITTED ON THIS FORM SINCE ANY CHANGES OR DELETIONS IN THE SPECIFICATIONS MAY RENDER YOUR BID/OFFER NON-RESPONSIVE IN WHICH CASE IT CANNOT BE CONSIDERED FOR AWARD.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

GG-B-00406b(DSA-DM) FSN:

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64, WHICH MAY BE USED.

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